

CreditGuard AI — Loan Risk Prediction System

Review 2 Documentation | 18-Feature XGBoost ML Pipeline & System Architecture

1. Executive Summary

For the **2nd Review**, the CreditGuard AI loan risk assessment system has been upgraded from a 13-feature baseline model to a comprehensive **18-feature Machine Learning pipeline**. The primary production classifier model is **XGBoost Classifier** (deployed on FastAPI backend & live cloud server), evaluated side-by-side against a **Random Forest Classifier** baseline in Jupyter Notebook. The web application features an expanded 18-input evaluation form, automated risk recommendation engine (loan capping, EMI calculation, driver flags), SQLite database persistence, mobile responsive design, and live cloud deployment on Render.

2. The 5 New Features Added for Review 2

The dataset (Dataset_XGBoost.csv) and production system have been enhanced with **5 brand new demographic, financial, and asset backing features** to improve risk separation:

Feature Name	Data Type	Possible Values / Range	Financial & Risk Rationale
marital_status	Categorical	Single, Married, Divorced	Evaluates household stability and shared financial responsibility.
number_of_dependents	Numeric (Integer)	0 to 20 dependents	Measures fixed family expenditure obligations and disposable income pressure.
vehicle_ownership	Categorical	Yes (Owned), No	Acts as secondary unencumbered asset backing and collateral indicator.
bank_account_age	Numeric (Integer)	0 to 80 years	Measures customer banking relationship tenure and financial stability history.
savings_balance	Numeric (Float)	\$0 to \$500,000+	Determines immediate liquid cash reserves to cushion against unexpected defaults.

3. Machine Learning Model Performance (Review 2 Evaluation)

Both models were trained and calibrated using decision boundary thresholding to achieve optimal accuracy and zero-error execution:

Evaluation Metric	Random Forest Classifier	XGBoost Classifier (Primary Deployed)
Accuracy	96.20%	95.80%
Precision	87.74%	95.74%
Recall	96.47%	84.98%
F1 Score	91.90%	90.04%
ROC AUC	0.9927	0.9206

4. Full 18-Feature System Schema Breakdown

Category	Feature Field Name	Description & Constraints
Demographics	person_age, person_gender, person_education, marital_status*, number_of_dependents*	Age (18-100), Gender, Higher Education Level, Marital Status, Household Dependents.

Financial & Credit	person_income, person_emp_exp, person_home_ownership, savings_balance*, bank_account_age*, credit_score, cb_person_cred_hist_length, previous_loan_defaults_on_file	Annual Income (\$), Work Exp (Yrs), Housing (RENT/OWN/MORTGAGE), Liquid Savings (\$), Bank Tenure (Yrs), FICO (300-850), Credit Hist (Yrs), Prior Default Flag.
Loan Proposal	loan_amnt, loan_intent, loan_int_rate, loan_percent_income	Requested Loan (\$), Purpose (EDUCATION, MEDICAL, PERSONAL, DEBT), Rate (%), Debt-to-Income Ratio (%).

** Denotes new features introduced in Review 2.*

5. System Architecture & Deployment Details

- **Frontend Architecture:** React (Vite/Webpack) SPA with responsive grid layout, off-canvas mobile navigation drawer, real-time risk gauge rendering, and 18-input assessment forms.
- **Backend Architecture:** FastAPI asynchronous REST server loading pre-compiled XGBoost pipeline transformers, auto-migrating SQLite database (loans.db), and dynamic risk factor analysis.
- **Live Cloud URL:** <https://loan-risk-prediction-z9jk.onrender.com>
- **Database Sync Script:** Includes python backend/live.py with 60-second auto-retry to sync cloud server records with local VS Code SQLite viewer.